■■ Doherty Threshold

■ What is it?

The point where even a small delay makes people lose interest.

Research in the 1970s found that computer interactions need to respond within **400 milliseconds** (less than half a second). Beyond this, people feel like they are waiting, which breaks the flow.

For example, think about waiting for a web page to load—how long before you give up and close it?

■■ History

The **Doherty Threshold** was discovered in the **1970s by Walter Doherty and Arvind Thadani** while studying human-computer interactions.

They showed that when systems respond within 400 ms, users feel more engaged and productive.

Fast forward to today: with an online attention span averaging **8.25 seconds (2024)**, speed matters more than ever.

■ The Psychology Behind It

- **Fast responses** keep people engaged in “flow.”
- **Delays** force users to stop and think, breaking concentration.
- Longer waits = frustration, drop-offs, and lost productivity.

This threshold explains why modern apps and sites invest heavily in speed and smooth performance.

■■ Why It Matters

Delays beyond the threshold can cause:
- Lower interest and engagement
- Higher bounce rates
- Drop in conversions

- Frustrated, impatient users
- Reduced quality of interaction

■ How to Apply It

- **Prioritize speed** → fast servers, optimized databases, minified code.
- **Reduce waiting time** → compress images, streamline processes.
- **Add feedback** → loading indicators, animations, or progress bars.
- **Refocus users** → offer small distractions (like games or tips) during unavoidable waits.

■ Theory in Action

- **Chrome browser** shows a dinosaur mini-game when offline, keeping users engaged.
- **Airport self-check-in kiosks** speed up boarding by reducing long waiting lines.

■ Final Thought

Users expect **instant responses**. Even tiny delays can break focus and push them away.

Design with respect for people’s time—whether by improving speed or adding clever distractions. Happy users are engaged users.